



Snowflake Enablement: Cortex Code

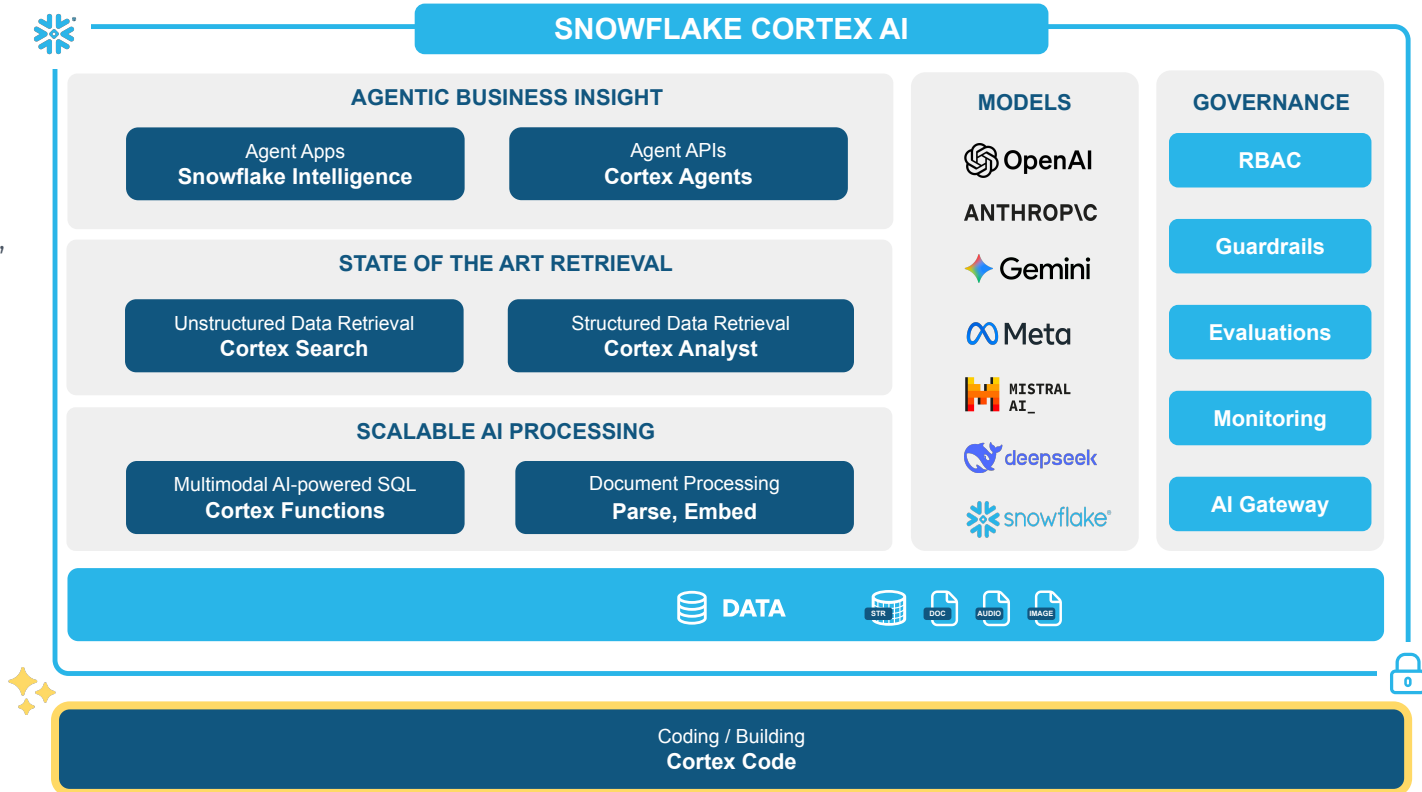
April 2026



Native AI & Agents Inside Snowflake

Cortex Code

AI coding agent that generates and iterates on Snowflake-native code (SQL, Python, Streamlit, etc.) grounded in your data and environment.



Cortex Code: AI Coding Agent for Data Teams

THE CURRENT STATE

Traditional Dev Friction

FRAGMENTED TOOLCHAIN

Workflows spanning dbt, Airflow, SQL, and Notebooks → constant context switching.

SLOW ITERATION

Manual refactoring and tribal knowledge slow debugging and deployments.

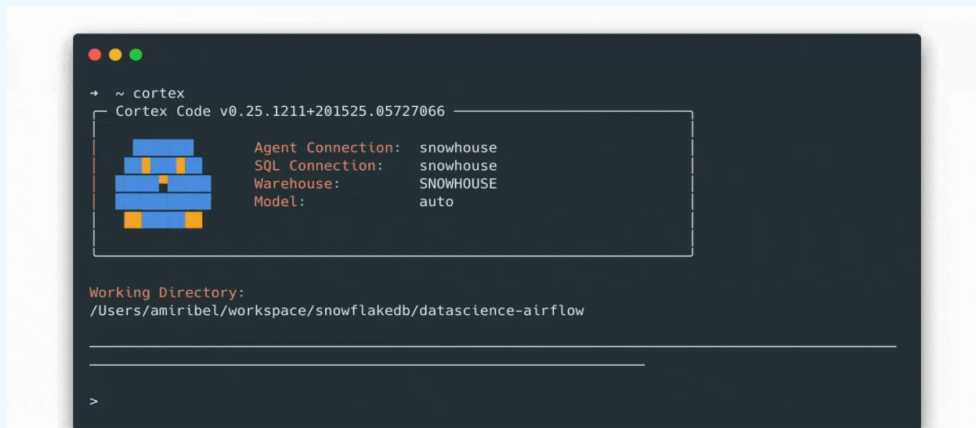
GOVERNANCE FRICTION

Access controls and policies bottleneck rapid prototyping.

THE SOLUTION

AI-Powered Development

From natural language to production-ready code across your entire data stack.
CLI & in Snowsight.



The Right Tool for Data Teams

An AI Coding Agent built to get you to production faster

UNDERSTANDS SNOWFLAKE



A built-in expert on Snowflake, it has up-to-the-minute knowledge of your specific data catalog: databases, schemas, semantic models and more.

INTELLIGENCE

CONTEXT AWARE



The agent knows where you are in the UI, what data you're working with, and retains your conversational history as you navigate.

RELEVANCE

AUTOMATES COMPLEX TASKS



Integrates natively with your stack. dbt, Git, SQL, Python, and Streamlit. No changes to how you work.

INTEGRATION

SECURE BY DESIGN



Cortex Code uses Snowflake RBAC, ensuring it only sees and acts on the data and objects you have access to

GOVERNANCE



Accelerate Your Data and AI Workflows



CORTEX CODE



DATA DISCOVERY

Search the catalog for data with semantic understanding



ADMIN TASK

Manage users, permissions, costs and governance



DATA ENGINEERING

Build & optimize pipelines with dbt and AirFlow



AI & AGENTS

Build, evaluate, optimize, debug agents and semantic views



ML & DATA SCIENCE

Auto-generate ML pipelines and feature engineering



ADVANCED ANALYTICS

Complex Analysis with Python and SQL generation



BI & APPS

Streamlit apps, dashboards & data applications



CODE MIGRATION & REFACTORING

Refactor and convert code from other platforms

**UNDERSTANDS
SNOWFLAKE & DATA**

**AUTOMATES
COMPLEX TASKS**

**CONTEXT AWARE
CLI & UI**

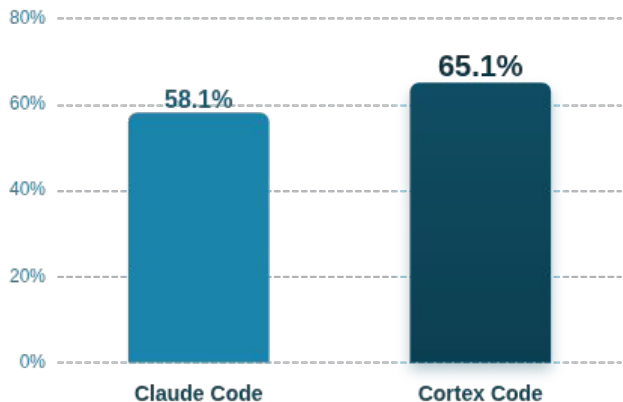
**SECURE BY
DESIGN**



Cortex Code Is the World's Best Data Coding Agent

Pass Rates on ADE-Bench

Higher is better (43 total tasks)



Model Version: Opus 4.6

BENCHMARK RESULTS

Cortex Code achieves comparable rates but demonstrates superior execution efficiency, with a **15.7% higher efficiency score** compared to Claude Code.

TOOL CALLS

854 ↓ 48%

vs 1,657 (Claude)

AVG STEPS

19.9 ↓ ~50%

vs 38.5 (Claude)

OPERATIONS

File Reads **2x Fewer**
Bash Cmds **4x Fewer**

Why Cortex Code is More Efficient

TARGETED VS. EXHAUSTED

Cortex performs targeted exploration rather than exhaustive scanning, significantly reducing IO operations.

SQL-NATIVE APPROACH

Utilizes **snowflake_sql_execute** for native execution, avoiding heavy reliance on bash tools common in Claude's workflow.



Snow-on-Snow: 10x Faster Root Cause Analysis for CX

10X

Faster Resolution

BEFORE
1 Hour to Days

AFTER
15 mins to 2 hrs

JIRA QUALITY

"Help us debug" → "Confirm this fix"

ENG EFFORT

Reduced from days to minutes

Context | Getting the right context together

Semantic Views

GET_ENVOY_LOGS, GET_JOBS for correct query results

Tool Integration

MCP connections to 10+ systems (Snowflake, SFDC, JIRA, GitHub, Glean, Confluence, etc.)

Intelligence | Data agent with encoded investigation patterns

Skills & Runbook

Reusable prompts encoding investigation patterns ("How do I debug throttling?")

Planning & Reasoning

Dynamic investigation paths & anomaly detection

Interface | Surfaced to Engineers in Cortex Code

Profiles

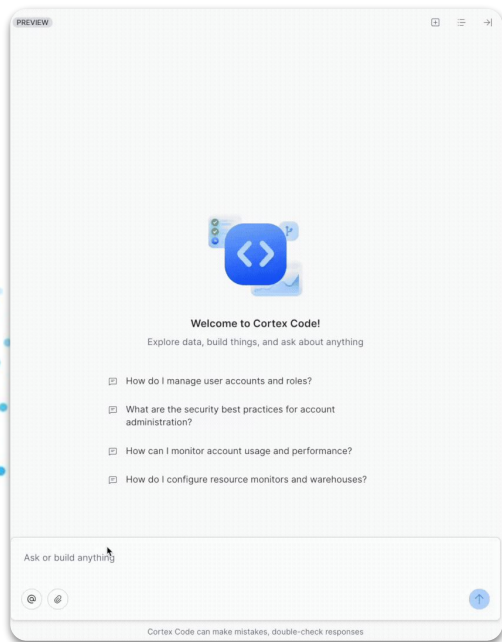
Packaged configurations (skills + tools + settings) stored centrally

No Setup Required

Any engineer gets the same toolset instantly available



Where to Access Cortex Code



Snowsight



CLI



BUILDING WITH BEST PRACTICES GOVERNANCE



Governance - Key Pillars

ACCESS MANAGEMENT

Fine-grained access control to ensure no data, agents, or other artifacts fall into the wrong hands.

> What tables are available in the CUSTOMER_DATA db

* Let me search for tables in the CUSTOMER_DATA database.

✗ **SNOWFLAKE SQL EXECUTE** List all tables in CUSTOMER_DATA database
└ Error executing SQL: SQL compilation error:

✓ **SNOWFLAKE SQL EXECUTE** Search for CUSTOMER_DATA database
0 row(s) affected.

* The CUSTOMER_DATA database doesn't exist or your current role doesn't have access to it.

Let me check what databases are available:

PERMISSIONING MODES

Confirm, Plan, and Bypass modes allow for precise control of local and remote operations.

> Create a new folder in my root directory called test

✓ **BASH** ls ~/test 2>/dev/null; echo "EXIT:\$?"
└ EXIT:1

Execute Command [HIGH]

mkdir ~/test

- o 1. Yes
 - o 2. Allow this specific command in the future
 - o 3. Allow this specific command for this session
 - o 4. Always allow using 'mkdir'
 - o 5. Allow using 'mkdir' for this session
 - > o 6. No (tab to add instructions)
- ↵ enter 1-6 esc

MODEL RBAC

Control which models are available account wide or to individual Cortex Code Users. Leverage Auto mode to pick best model available.

Select a model

- > o Auto – Picks best available model for your account/region (current)
- o Claude Opus 4.6
 - o Claude Opus 4.5
 - o Claude Sonnet 4.6
 - o Claude Sonnet 4.5
 - o Claude Sonnet 4.0
 - o GPT-5.2
- ↵ enter esc



Permission Modes Deep Dive

Confirm Mode (default)

Prompts the user before file edits, bash commands, etc. All proposed operations will classify risk level to user for approval.

```
> Create a new folder in my root directory called test
```

```
✓ BASH ls ~/test 2>/dev/null; echo "EXIT:$?"  
  EXIT:1
```

Execute Command [HIGH]
mkdir ~/test

- 1. Yes
 - 2. Allow this specific command in the future
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Plan Mode (/plan)

Agent polls user for additional detail and proposes a plan but doesn't execute any action until approved.

```
> create a script to clean up tables in snowflake  
* Let me clarify what you're looking for before planning.
```

```
✓ ASK_USER_QUESTION
```

```
✓ ASK_USER_QUESTION
```

Proposed plan:

```
## Plan: Empty Table Report for MARKETING_CAMPAIGNS
```

Create a single SQL script that:

1. **Query** 'INFORMATION_SCHEMA.TABLES' in the 'MARKETING_CAMPAIGNS' database to find all tables with 'ROW_COUNT' = 0
2. **Report columns**: database name, schema name, table name, table type, creation time, last altered time, and bytes (to confirm truly empty)
3. **Sort** by schema then table name for easy review
4. **Exclude** 'INFORMATION_SCHEMA' itself from results

Output: One SQL query, executed directly against your Snowflake connection ('EBOTWICK_DEMO').

Plan Confirmation

Would you like Cortex to proceed with this plan?

- > ○ Yes (tab to add instructions)
 - Yes, in background
 - No

☐ Clear context before executing (shift+tab to toggle)

Clears the conversation before the plan was generated so the agent can better focus on the plan
↑ enter esc

⚠ Bypass Mode (/bypass)

Auto-approves all tool without polling user (not recommended - can be disabled via managed settings)

> Bypass safeguards mode enabled
All tool calls will be auto-approved without confirmation.
Use /bypass-off to disable.

> Be careful!

>> **bypass safeguards** (shift+tab) | **plan mode: off** (ctrl+p)
view: compact (ctrl+o to cycle) * ? for help



Model RBAC Deep Dive

Model Governance

Account-Level Model Allowlist

An ACCOUNTADMIN sets the **CORTEX_MODELS_ALLOWLIST** parameter to explicitly permit or block models account-wide. Setting it to 'None' blocks ALL model access unless explicitly re-enabled.

Per-Model RBAC (Fine-Grained Control):

Each Cortex model has its own application role in the **SNOWFLAKE.MODELS** schema (e.g., **CORTEX-MODEL-ROLE-CLAUDE-SONNET-4-5**). Grant or revoke access per role for explicit, auditable control.

Model Selection in Cortex Code

/model

In-session command to select from available models. Model selection applies only to current session.

CORTEX_AGENT_MODEL

Environment variable to set for model selection. Can also be included in settings.json like below.

```
{"env": {"CORTEX_AGENT_MODEL":  
"claude-sonnet-4-5"}}
```

cortex -- model

CLI Flag to specify model to be used in a specific session



Governance - Configuration Paths

Settings.json

Multiple layers of enforceable settings at the account or individual level to ensure company wide compliance and individual optimizations

```
{
  "version": "1.0",
  "permissions": {
    "onlyAllow": ["account(mycompany-prod)", "account(mycompany-staging)"],
    "deny": ["mcp_*"],
    "defaultMode": "allow",
    "dangerouslyAllowAll": false
  },
  "settings": {
    "forceNoHistoryMode": false,
    "forceSandboxEnabled": true,
    "forceSandboxMode": "regular"
  },
  "required": {
    "minimumVersion": "1.0.0"
  },
  "defaults": {
    "connectionName": "prod",
    "profileName": "corporate",
    "theme": "dark"
  },
  "ui": {
    "showManagedBanner": true,
    "bannerText": "[Secure] Managed by Corporate IT",
    "hideDangerousOptions": true
  }
}
```

Hooks.json

hooks.json allows for customizable controls for allowing or preventing granular operations

```
Block destructive SQL (DROP, DELETE, TRUNCATE):
{
  "hooks": {
    "preToolUse": [
      {
        "matcher": "snowflake_sql_execute",
        "hooks": [
          {
            "type": "command", "command": "/path/to/block-destructive-sql.sh", "timeout": 10
          }
        ]
      }
    ]
  }
}

Audit all Bash commands:
{
  "hooks": {
    "postToolUse": [
      {
        "matcher": "Bash",
        "hooks": [
          {
            "type": "command", "command": "/path/to/audit-logger.sh", "timeout": 10
          }
        ]
      }
    ]
  }
}

Block all file writes outside a project directory:
{
  "hooks": {
    "preToolUse": [
      {
        "matcher": "write|edit",
        "hooks": [
          {
            "type": "command", "command": "/path/to/restrict-writes.sh", "timeout": 10
          }
        ]
      }
    ]
  }
}
```

Connections.toml

Define connection profiles to various snowflake environments and configurations

> What connections do I have available

```
✓ BASH cortex connections list
- {
  "active_connection": "EBOTWICK_DEMO",
  ... (23 more lines)
```

*

Connection	Account	User	Role
EBOTWICK_DEMO ...	sfse-northamerica-demo...	EBOT...	ACCOUNTADM...
DEMO_ANALYST	sfse-northamerica-demo...	EBOT...	DATA_ANALY...
snowhouse	SFCOGSOPS-SNOWHOUSE_AW...	EBOT...	ENGINEER
JAN26_TRIAL	EHDRLVK-BAB26722	EBOT...	ACCOUNTADM...

Currently using **EBOTWICK_DEMO**. Switch with `cortex connections set <name>` or `/connection <name>`.



Governance - Best Practices

Enable Plan Mode - `/plan` - review actions before execution

Use Scoped Roles - Avoid using overly-permissioned roles with Cortex Code

Disable Bypass Mode - Managed settings: `"dangerouslyAllowAll": false`

Deploy Hooks - Pre-execution hooks for production-specific policies

Force Sandbox - Via managed settings

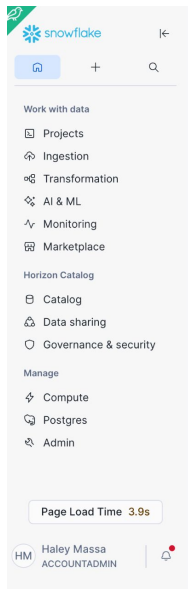
Monitor Usage - Review `SNOWFLAKE.ACCOUNT_USAGE.CORTEX_CODE_CLI_USAGE_HISTORY` regularly



CREATING COCO CLI CONNECTION



Validate Cortex Code in Snowsight



Home

Search this account, Marketplace, and documentation

Quick actions



Query data

Create a SQL file in Workspaces



Create User

Provide access to collaborators



Create Warehouse

Manage compute resources



Upload local files

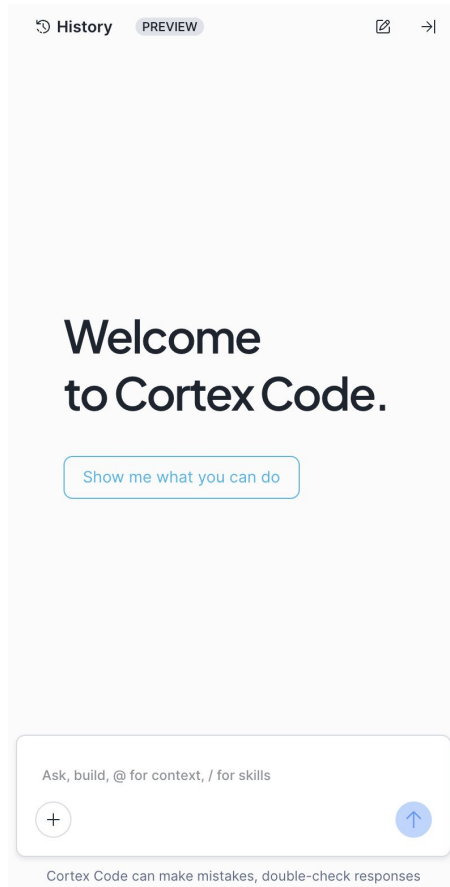
Quickly convert data into tables

Recently viewed

[All projects](#) [Files](#) [Notebooks](#) [Streamlit](#) [Dashboards](#) [Folders](#)

TITLE	TYPE	VIEWS ↓
02_b_ManyModelTraining_Wafer	Notebook	2 days ago
02_c_DistributedTraining_DDP	Notebook	2 days ago
Untitled 8.sql	SQL File	3 days ago
Untitled 9.sql	SQL File	4 days ago
Untitled 13.sql	SQL File	1 week ago
Agent Permission Generator	Streamlit	2 weeks ago

Access Cortex Code via Snowsight by clicking on the blue star in the bottom right corner in the UI



Installing the CoCo CLI on Mac

INSTALL VIA TERMINAL

```
curl -LsS https://ai.snowflake.com/static/cc-scripts/install.sh | sh
```

Architectures

- x64 – Supported.
- ARM64 – Supported

VERIFY INSTALL

```
cortex --version
```

During Installation

- When prompted to add cortex to your PATH, respond with y.
- If the cortex command is not recognized after installation:
- Close and reopen your terminal, or
- Restart your machine to ensure PATH changes take effect.



Installing the CoCo CLI on Mac

CONFIGURE SNOWFLAKE CONNECTIONS

- Cortex Code uses the same connection files as SnowCLI, under your home directory. On Macs that's typically:
- `~/snowflake/config.toml` (or `connections.toml` if you already use that).
- Minimal example (adapt to your accounts/roles):

```
default_connection_name = "DEMO" # NAME FOR DEFAULT SNOWFLAKE CONNECTION
[connections.DEMO] # NAME FOR SNOWFLAKE CONNECTION (SAME AS ABOVE IF DEFAULT)
account = "<YOUR_DEMO_ACCOUNT>" # e.g. ACCOUNT IDENTIFIER
user = "<YOUR_DEMO_USERNAME>" # e.g. AUTHENTICATION VARIABLE
password = "<YOUR_DEMO_PAT>"
role = "<YOUR_DEMO_ROLE>" # e.g. NEED A ROLE FOR CONNECTION
warehouse = "<YOUR_DEMO_WAREHOUSE>"
```

LAUNCH CORTX CODE CLI USING YOUR SQL CONNECTION

```
cortex -c DEMO # connect to connection name
```



Installing the CoCo CLI on Windows

INSTALL VIA POWERSHELL

```
irm https://ai.snowflake.com/static/cc-scripts/install.ps1 | iex
```

Architectures

- x64 – Supported.
- ARM64 – Support is in progress. Current builds may rely on experimental runtime support and are not yet recommended for production use.

VERIFY INSTALL

```
cortex --version
```

During Installation

- When prompted to add cortex to your PATH, respond with y.
- If the cortex command is not recognized after installation:
 - Close and reopen your terminal, or
 - Restart your machine to ensure PATH changes take effect.



Installing the CoCo CLI on Windows

CONFIGURE SNOWFLAKE CONNECTIONS

- Cortex Code uses the same connection files as SnowCLI, under your home directory. On Windows that's typically:
- `%USERPROFILE%.snowflake\config.toml` (or `connections.toml` if you already use that).
- Minimal example (adapt to your accounts/roles):

```
default_connection_name = "DEMO" # NAME FOR DEFAULT SNOWFLAKE CONNECTION
[connections.DEMO] # NAME FOR SNOWFLAKE CONNECTION (SAME AS ABOVE IF DEFAULT)
account = "<YOUR_DEMO_ACCOUNT>" # e.g. ACCOUNT IDENTIFIER
user = "<YOUR_DEMO_USERNAME>" # e.g. AUTHENTICATION VARIABLE
password = "<YOUR_DEMO_PAT>"
role = "<YOUR_DEMO_ROLE>" # e.g. NEED A ROLE FOR CONNECTION
warehouse = "<YOUR_DEMO_WAREHOUSE>"
```

LAUNCH CORTX CODE CLI USING YOUR SQL CONNECTION

```
cortex -c DEMO # connect to connection name
```



Creating a Snowflake Connections Config

ON MAC

Run this code in your terminal to create the connections config file

```
mkdir -p ~/.snowflake  
touch ~/.snowflake/config.toml  
chmod 600 ~/.snowflake/config.toml
```

Then run this command to open the file to edit and add in config

```
open -e ~/.snowflake/config.toml
```

ON WINDOWS

Run this code in powershell to create the connection config file

```
mkdir $env:USERPROFILE\.snowflake -Force  
New-Item -ItemType File -Path $env:USERPROFILE\.snowflake\config.toml  
-Force
```

Then run this command to open the file to edit and add in config

```
notepad $env:USERPROFILE\.snowflake\config.toml
```

Filling out the Default Connection

LOG IN AND FIND SETTING INFORMATION

The screenshot shows the Snowflake web interface. In the top left, the user 'Haley Massa' is logged in as 'ACCOUNTADMIN'. The account 'HTB43887' is selected. The account details modal is open, showing the account name 'JNWOMPH', the account identifier 'HTB43887', and the role 'ACCOUNTADMIN'.

CLICK ON CONNECTION AND FIND DEFAULTS

Account Details		
Account Config File Connectors/Drivers SQL Commands		
NAME	VALUE	
Account identifier ⓘ	JNWOMPH-HTB43887	📄
Data sharing account identifier ⓘ	JNWOMPH.HTB43887	📄
Organization name	JNWOMPH	📄
Account name	HTB43887	📄
Account/Server URL	JNWOMPH-HTB43887.snowflakecomputing.com	📄
Login name ⓘ	HMASSA	📄
Role	ACCOUNTADMIN	📄
Account locator	NIB84296	ACCOUNTADMIN 📄
Cloud platform	AWS	📄
Edition	Enterprise	📄



Ways to Authenticate Connections

CREATE AND AUTHENTICATE A CONNECTION TO SNOWFLAKE

There are many ways to create a connection to Snowflake. These are included into your connection configuration as credentials to identify you as logging into your connection. Here are some examples of environment variables used in Snowflake credentials:

- `SNOWFLAKE_ACCOUNT`
- `SNOWFLAKE_USER`
- `SNOWFLAKE_PASSWORD`
- `SNOWFLAKE_TOKEN`
- `SNOWFLAKE_TOKEN_FILE_PATH`
- `SNOWFLAKE_OAUTH_CLIENT_ID`
- `. . .`

Find more examples and walk through creating your connection parameter in the [Manage Snowflake connections guide](#).

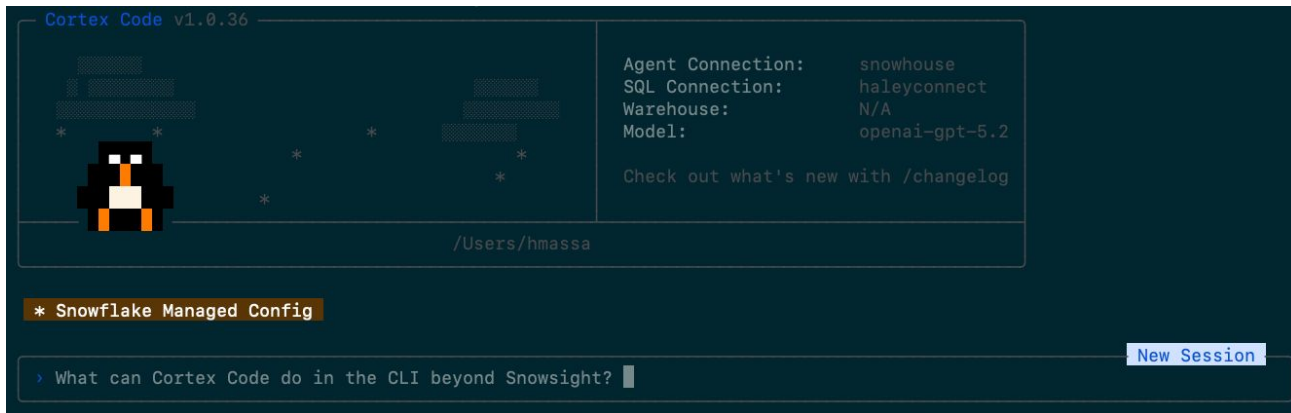


Launching Cortex Code

LAUNCH CORTEx CODE CLI USING YOUR SQL CONNECTION

```
cortex -c DEMO # connect to connection name
```

- After connecting to cortex code (using the connect -c command and your connection name (in this example, DEMO), you will see the Cortex Code CLI interface
- From here is where we will cover more advanced usages of Cortex Code in our session, but we can test the connection first by asking a simple question:



GETTING STARTED WITH CORTEX CODE



How Cortex Code Works

Step 1: Install

Install the cortex CLI

Step 2: Connect

Connect to your
Snowflake account
(SSO, key-pair, or
password)

Step 3: Ask

Start giving tasks in
natural language

Key Interaction Patterns

#DB.SCHEMA.TABLE — auto-injects table metadata and sample rows into your prompt

/sql SELECT ... — run SQL inline and see results instantly

\$skill-name — activate a specialized skill (e.g., \$cost-intelligence)

@path/to/file — include file context in your conversation

Cortex Code CLI

Contains bundled tools and skills and possibility to extend further

Cortex Code CLI contains **tools** and bundled **skills** specific to Snowflake capabilities. Is as well **extendable** via **custom skills** and **MCP** servers.

Tools:

- Plan Mode
- SQL Execution
- Snowflake Documentation Access
- Local files access
- Terminal / Shell commands
- Web Search
- Browser
- Model Context Protocol (MCP)
- git Support
- dbt Support
- Skills (bundled and custom)
- Agents.md
- Hooks

Skills & Plugins (bundled):

Migration & Assessment

- SnowConvert Assessment

Data Engineering & Pipelines

- dbt on Snowflake
- Dynamic Tables
- Openflow

Security, Governance & Compliance

- Data Governance
- Data Policy
- Data Classification
- Cost Management

AI & Machine Learning

- Machine Learning
- Agent Optimization
- Semantic Views

Applications & Visualization

- Build React App
- Streamlit in Snowflake
- Deploy to SPCS
- Snowflake Postgres

Platform Operations

- Skill Development
- Cortex Code Guide



Essential Shortcuts & Commands

KEYBOARD SHORTCUTS

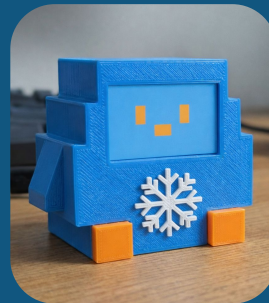
Shortcut	Action
Enter	Submit message
Ctrl+J	Insert newline
Ctrl+C	Cancel/Exit
Shift+Tab	Cycle modes (Confirm/Plan/Bypass)
Ctrl+P	Toggle Plan mode
Ctrl+D	Open full screen task view
Ctrl+B	Background bash process
Ctrl+O	Cycle display mode
Ctrl+V	Paste (supports images)

CONTEXT SHORTCUTS

Shortcut	Action	Example
@	File reference	@src/app.py
@file\$10-50	Specific lines	@config.yaml\$1-20
\$	Invoke skill	\$semantic-view-optimization
#	Snowflake table	#DB.SCHEMA.TABLE
!	Run bash command	!git status
/	Slash commands	/help, /sql, /plan
?	Quick help	?



CORTEX CODE WORKSHOP



Build A Snow Day Learning Fundamentals

Cortex Code Fundamentals

- ☐ Understand what Cortex Code is and how it differs from traditional development
- ☐ Access Cortex Code via Snowsight (UI)
- ☐ Configure a connection to run Cortex Code locally (CLI)
- ☐ See how Cortex Code understands your Snowflake environment (schema discovery)
- ☐ Execute SQL and workflows as yourself (your identity, your permissions)

Building with Cortex Code

- ☐ Build production-ready data pipelines in Snowflake
- ☐ Connect to external data/files and bring context into Snowflake
- ☐ Use bundled skills (\$dynamic-tables, \$semantic-view, etc)
- ☐ Create your own reusable skills for your team's workloads



Demo One: Pipeline Builder

From scattered ERP sources to a governed, chained Dynamic Table pipeline

The Story

Your company runs two ERP systems: SAP for manufacturing in Germany and Oracle for operations in the US. Both generate AP invoices, but with completely different column names and conventions.

Finance needs a single, consolidated view of all payables for reporting. Today this is done with manual SQL scripts that break every time a column changes.

You'll use Cortex Code to build a declarative pipeline using Dynamic Tables — Bronze to Silver (consolidation) to Gold (aggregation) — with automatic refresh and no orchestration code.

Skills Used

`$dynamic-tables`

4 demos • *Builds the foundation for all subsequent vignettes*

Demo 1.1 — Setup: Explore the source data

Demo 1.2 — Compare SAP and Oracle invoice schemas

Demo 1.3 — Generate Silver Dynamic Table

Demo 1.4 — Use the `$dynamic-tables` skill

Demo 1.5 — Save one proof query

What You'll Learn

- Ground CoCo on live schema context
- Generate consolidation DDL from cross-source metadata
- Generate downstream aggregation DDL + verification queries
- Save the workflow as a team skill (inputs / outputs)



Execution Modes in CLI

A look back at how we can run our different permissions:

Confirm Mode (default)

Prompts the user before file edits, bash commands, etc. All proposed operations will classify risk level to user for approval.

```
> Create a new folder in my root directory called test
```

```
✓ BASH ls ~/test 2>/dev/null; echo "EXIT:$?"  
  _ EXIT:1
```

Execute Command [HIGH]

```
mkdir ~/test
```

- o 1. Yes
 - o 2. Allow this specific command in the future
 - o 3. Allow this specific command for this session
 - o 4. Always allow using 'mkdir'
 - o 5. Allow using 'mkdir' for this session
 - o 6. No (tab to add instructions)
- ```
↑ enter 1-6 esc
```

## Plan Mode (/plan)

Agent polls user for additional detail and proposes a plan but doesn't execute any action until approved.

```
> create a script to clean up tables in snowflake
* Let me clarify what you're looking for before planning.
```

```
✓ ASK_USER_QUESTION
✓ ASK_USER_QUESTION
```

Proposed plan:

```
Plan: Empty Table Report for MARKETING_CAMPAIGNS
```

Create a single SQL script that:

1. `Query 'INFORMATION_SCHEMA.TABLES'` in the 'MARKETING\_CAMPAIGNS' database to find all tables with 'ROW\_COUNT' = 0
2. `Report columns`: database name, schema name, table name, table type, creation time, last altered time, and bytes (to confirm truly empty)
3. `Sort` by schema then table name for easy review
4. `Exclude` 'INFORMATION\_SCHEMA' itself from results

Output: One SQL query, executed directly against your Snowflake connection ('EBOTWICK\_DEMO').

### Plan Confirmation

Would you like Cortex to proceed with this plan?

- o Yes (tab to add instructions)
- o Yes, in background
- o No

Clear context before executing (shift+tab to toggle)

Clears the conversation before the plan was generated so the agent can better focus on the plan

```
↑ enter esc
```



## Bypass Mode (/bypass)

Auto-approves all tool without polling user (not recommended - can be disabled via managed settings)

> Bypass safeguards mode enabled  
All tool calls will be auto-approved without confirmation.  
Use /bypass-off to disable.

> Be careful!

>> **bypass safeguards** (shift+tab) | plan mode: off (ctrl+p)  
view: **compact** (ctrl+o to cycle) • ? for help





# Bundled Skills in Snowflake



# Pipeline Maintainer

Diagnose a stale pipeline, fix it, and onboard new sources

## The Story

It's a month after launch. The Silver pipeline from Vignette 1 is live, but something's wrong — data looks stale and stakeholders are asking questions.

Meanwhile, the Finance Transformation PMO just sent over an Excel file: they want to onboard two more ERP systems (Baan from EMEA and Workday from Americas). The spreadsheet has column mappings, business rules, and open questions.

You need to diagnose the issue, translate those business requirements into technical specs, evolve the pipeline to handle 4 sources, and set up monitoring so this doesn't happen again.

## Skills Used

## \$skill-development

5 demos • Requires Vignette 1 completed • Uses local (or staged) Excel file

Demo 2.1 — Read the local PRD

Demo 2.2 — Scaffolding a Custom Skill

Demo 2.3 — Run the PRD Evaluator Skill

Demo 2.4 — Apply the changes

Demo 2.5 — (optional) Save handoff materials

## What You'll Learn

- Turn an Excel requirements doc into executable change steps
- Capture the workflow as a reusable team skill/runbook
- Run our local skill
- Save materials to share updates with team members

# Build First Agent

Enable business users to ask natural language questions on our data

## The Story

The AP consolidation pipeline is running smoothly with 4 source systems. But only engineers can query it — business users can't write SQL.

You'll create a semantic view that translates the Silver table into business-friendly dimensions and measures, then wrap it in a Cortex Agent that answers natural language questions about AP invoices.

But an agent is only as good as its answers. You'll build an evaluation framework — test questions, logged responses, and human feedback — then use that data to audit and improve the agent's semantic view iteratively.

## What You'll Learn

- Generate a semantic view from live table metadata
- Create an agent with verifiable responses (answer + SQL + assumptions)
- Run evaluation on stakeholder created first golden dataset
- Optimize Data Agent to learn from wrong responses

## Skills Used

\$semantic-view

\$cortex-agent

4 demos • Requires Vignettes 1-2 completed • Introduces AI Layer

Demo 3.1 - Create Semantic View on Silver Data

Demo 3.2 - Create a Cortex Agent grounded on the semantic view

Demo 3.3 - Evaluate against a business user curated dataset

Demo 3.4 - Iterate on Agent from evaluation feedback



# Core Capabilities in \$Semantic-View-Optimize



## CREATE

Create new semantic views from scratch



## AUDIT

Audit semantic view, VQR testing, best practices, evaluation



## DEBUG

Analyzing errors, fixing failures, root cause analysis for specific SQL queries

My query keeps failing



Uses Debug Mode

Create a semantic view for my sales table



Uses Creation Mode

Test all my VQRs still work after schema changes



Uses Audit Mode VQR testing

Why does “revenue by region” generate the wrong SQL



Uses Debug Mode

Add metrics and filters to improve query success



Uses Audit Mode



# Core Capabilities in \$Agent-Optimize



## CREATE

Building new Cortex Agents



## ADHOC TEST

Test questions, and build evaluation datasets



## DEBUG QUERY

Debug specific query failures



## OPTIMIZE

Improve agent performance for production

Why does “show me last quarter’s revenue” fail



Uses Debug Mode

Build customer support agent to query data + respond



Uses Creation Mode

I want to test on 20 common questions



Uses Audit Mode

Why does “revenue by region” generate the wrong SQL



Uses Debug Mode

Agent only works 60% accuracy on my test set



Uses Optimize





# APPENDIX



# How do we evaluate an agent?

Our bundled Agent Optimization Skill supports 4-evaluation approaches:



## Ad-Hoc Testing

Interactively ask questions and manually review responses to explore agent behavior



## Curated Dataset Testing

Run business stakeholder provider questions with expected answers; score with LLM judge



## Feedback Based Testing

Review production queries in a Streamlit application and annotate responses



## Evaluation Dataset Curation

Build a representative question set from production logs or create from scratch

### This demo uses approach #2 : Curated Dataset Testing

- Business stakeholders provided 15-20 realistic questions with expected answers
- This collaboration is a common starting point for teams building out with Agents





# Settings Deep Dive

## Two Layers of Control

### User settings:

`~/.snowflake/cortex/settings.json`

(controlled by individual user)

### Managed settings:

`managed-settings.json`

(controlled by IT admin via MDM/SCCM)

*Managed settings override user settings. Users cannot bypass admin policy.*

### Settings precedence (highest → lowest)

1. Managed settings (admin policy)
2. Session commands (`/plan`, `/bypass`)
3. CLI arguments
4. Environment variables
5. User config files
6. Defaults

## What Admins Can Enforce

**Permissions:** Tool allowlists/denylists, disable bypass, restrict Snowflake accounts

**Settings:** Force no history, force sandbox enabled

**Required:** Enforce minimum CLI version

**Defaults:** Default connection, profile, theme

**UI:** Managed banner, hide dangerous options



# Hooks Deep Dive

## What it does & how it works

**What it does:** Custom pre/post-execution rules that auto-approve or block specific operations.

### How it works:

- **hooks.json** defines rules triggered before or after tool calls
- Matcher targets a specific tool (Bash, Write, SQL, etc.)
- Hook script returns allow, deny, or a custom reason
- Hooks are not bypassed even in bypass mode

**Customer takeaway:** *"Your security team writes the rules. CoCo enforces them automatically."*

**Example:** Auto-approve safe bash, block rm -rf:

```
{
 "hooks": {
 "PreToolUse": [
 {
 "matcher": "Bash",
 "hooks": [{
 "type": "command",
 "command": "bash hooks/policy.sh"
 }]
 }
]
 }
}
```

Hook script can inspect the command and return:

- **"permissionDecision": "allow"** → approved
- **"permissionDecision": "deny"** → blocked with reason



# Connections Deep Dive

## What it does & how it works

**What it does:** Named connection profiles scoping each session to a specific Snowflake environment, role, and warehouse.

### How it works:

- Defined in `~/.snowflake/connections.toml` (shared with Snowflake CLI)
- Each profile sets account, role, warehouse, database, schema
- Admins can restrict which accounts are connectable via managed settings

**Customer takeaway:** *"Users get the right access for the right environment. No accidental prod writes from a dev session."*

**Example:** Separate dev vs. prod profiles:

```
[dev]
account = "myco-dev"
role = "DEVELOPER"
warehouse = "DEV_WH"
database = "DEV_DB"
```

```
[prod_readonly]
account = "myco-prod"
role = "ANALYST"
warehouse = "REPORTING_WH"
database = "PROD_DB"
```

Launch with: `cortex -c prod_readonly`



# Governance - Roadmap

## What we're building

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### **AUDIT LOGGING & OBSERVABILITY DATA**

Prompt and response logging for analysis of usage patterns and performance insights

### **COST CONTROLS & BUDGETS**

Manage usage costs and prevent runaway costs on a user or account level

### **SUPPORT FOR CORTEX GUARDRAILS**

Protection against prompt injection and other LLM risks. Planned for Q2.

### **CORTEX CODE PROFILES**

Subsets of global configuration settings for skills, MCP servers, settings, hooks etc.  
Select a profile at session start to determine behavior for that session

